

2

Tensor Calculus

To describe the mechanical behavior of continuous media we will use scalars, vectors and second-order tensors which in general may vary from point to point in a material body. Since a body may be identified with a subset of Euclidean space we will be interested in functions of the form $\phi : \mathbf{E}^3 \rightarrow \mathbf{R}$, $\mathbf{v} : \mathbf{E}^3 \rightarrow \mathcal{V}$ and $\mathbf{S} : \mathbf{E}^3 \rightarrow \mathcal{V}^2$. Moreover, to describe the shapes of material bodies, and their reactions to changes in shape, we will be interested in functions of the form $\chi : \mathbf{E}^3 \rightarrow \mathbf{E}^3$, $g : \mathcal{V}^2 \rightarrow \mathbf{R}$ and $\mathbf{G} : \mathcal{V}^2 \rightarrow \mathcal{V}^2$.

In this chapter we study various calculus concepts for different types of functions on $\mathbf{E}^3$ and $\mathcal{V}^2$. The important ideas that we introduce are: (i) the derivative or gradient, divergence, curl and Laplacian for different types of functions on $\mathbf{E}^3$; (ii) the Divergence, Stokes' and Localization Theorems pertaining to integrals over different types of subsets of $\mathbf{E}^3$; (iii) the derivative of different types of functions on $\mathcal{V}^2$.

For all the definitions and results in this chapter we assume, without further explicit statement, that the functions being differentiated are smooth in the sense that partial derivatives of all orders exist and are continuous. This is always stronger than what we need, but allows for a clean presentation.

2.1 Preliminaries

2.1.1 Points, Tensors and Representations

Throughout the remainder of our developments we assume that a single, fixed coordinate frame $\{\mathbf{e}_i\}$ has been specified for Euclidean space $\mathbf{E}^3$. Under this assumption we may identify points $\mathbf{x} \in \mathbf{E}^3$, vectors $\mathbf{v} \in \mathcal{V}$ and second-order tensors $\mathbf{S} \in \mathcal{V}^2$ with their respective matrix

representations; in particular, we write

$$\mathbf{x} = x_i \mathbf{e}_i = \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} \in \mathbb{R}^3, \quad \mathbf{v} = v_i \mathbf{e}_i = \begin{Bmatrix} v_1 \\ v_2 \\ v_3 \end{Bmatrix} \in \mathbb{R}^3,$$

and

$$\mathbf{S} = S_{ij} \mathbf{e}_i \otimes \mathbf{e}_j = \begin{pmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

Thus $\mathbb{E}^3$ and $\mathcal{V}$ are both identified with $\mathbb{R}^3$, and $\mathcal{V}^2$ is identified with $\mathbb{R}^{3 \times 3}$. Any function of a point $\mathbf{x}$ is a function of the three real variables (x_1, x_2, x_3) , and any function of a second-order tensor $\mathbf{S}$ is a function of the nine real variables $(S_{11}, S_{12}, \dots, S_{33})$.

2.1.2 Standard Norms, Order Symbols

Here we exploit the above identifications to define norms on the spaces $\mathbb{E}^3$, $\mathcal{V}$ and $\mathcal{V}^2$ and introduce the order symbols $\mathcal{O}$ and o .

Definition 2.1 *By the standard (Euclidean) norms on the spaces $\mathbb{E}^3$, $\mathcal{V}$ and $\mathcal{V}^2$ we mean the scalar functions defined by*

$$\begin{aligned} |\mathbf{x}| &= \sqrt{\mathbf{x} \cdot \mathbf{x}}, & \forall \mathbf{x} \in \mathbb{E}^3, \\ |\mathbf{v}| &= \sqrt{\mathbf{v} \cdot \mathbf{v}}, & \forall \mathbf{v} \in \mathcal{V}, \\ |\mathbf{S}| &= \sqrt{\mathbf{S} : \mathbf{S}}, & \forall \mathbf{S} \in \mathcal{V}^2. \end{aligned}$$

□

In the same way that $|\mathbf{x}|$ and $|\mathbf{v}|$ provide a measure of the magnitude or size of the vectors $\mathbf{x}$ and $\mathbf{v}$, the quantity $|\mathbf{S}|$ provides a measure of the magnitude or size of $\mathbf{S}$. It has all the usual properties of a norm. Using these standard norms we can define limits in the spaces $\mathbb{E}^3$, $\mathcal{V}$ and $\mathcal{V}^2$, along with continuity of functions between them.

Definition 2.2 *Consider a function $\mathbf{f} : U \rightarrow W$, where U and W denote any of the spaces $\mathbb{E}^3$, $\mathcal{V}$, $\mathcal{V}^2$ or $\mathbb{R}$.*

(i) *If there are constants $C > 0$ and $r > 0$ such that $|\mathbf{f}(\mathbf{u})| \leq C|\mathbf{u}|^r$ as $\mathbf{u} \rightarrow \mathbf{0}$, then we write*

$$\mathbf{f}(\mathbf{u}) = \mathcal{O}(|\mathbf{u}|^r) \quad \text{as } \mathbf{u} \rightarrow \mathbf{0}.$$

(ii) If there is a constant $r > 0$ such that $|\mathbf{f}(\mathbf{u})|/|\mathbf{u}|^r \rightarrow 0$ as $\mathbf{u} \rightarrow \mathbf{0}$, then we write

$$\mathbf{f}(\mathbf{u}) = o(|\mathbf{u}|^r) \quad \text{as } \mathbf{u} \rightarrow \mathbf{0}.$$

The symbols $\mathcal{O}$ and o are called the **standard order symbols**. □

The order symbols provide a way to describe the behavior of a function $\mathbf{f}(\mathbf{u})$ as $\mathbf{u}$ approaches zero, or, by translation, any other given value. The statement $\mathbf{f}(\mathbf{u}) = \mathcal{O}(|\mathbf{u}|^r)$ as $\mathbf{u} \rightarrow \mathbf{0}$ means that $|\mathbf{f}(\mathbf{u})|$ goes to zero at least as fast as $|\mathbf{u}|^r$. The statement $\mathbf{f}(\mathbf{u}) = o(|\mathbf{u}|^r)$ as $\mathbf{u} \rightarrow \mathbf{0}$ means that $|\mathbf{f}(\mathbf{u})|$ goes to zero faster than $|\mathbf{u}|^r$. Thus $\mathbf{f}(\mathbf{u}) = o(|\mathbf{u}|^r)$ implies $\mathbf{f}(\mathbf{u}) = \mathcal{O}(|\mathbf{u}|^r)$, but the converse need not be true. However, when $p > r$, we find that $\mathbf{f}(\mathbf{u}) = \mathcal{O}(|\mathbf{u}|^p)$ implies $\mathbf{f}(\mathbf{u}) = o(|\mathbf{u}|^r)$.

Given two functions $\mathbf{f}, \mathbf{g} : U \rightarrow W$ we use the notation

$$\mathbf{f}(\mathbf{u}) = \mathbf{g}(\mathbf{u}) + \mathcal{O}(|\mathbf{u}|^r) \quad \text{as } \mathbf{u} \rightarrow \mathbf{0}$$

to signify that

$$\mathbf{f}(\mathbf{u}) - \mathbf{g}(\mathbf{u}) = \mathcal{O}(|\mathbf{u}|^r) \quad \text{as } \mathbf{u} \rightarrow \mathbf{0}.$$

Similar notation is also used for the order symbol o . We sometimes indicate the limit $\mathbf{u} \rightarrow \mathbf{0}$ by saying that $\mathbf{u}$ is small. When there is no cause for confusion we omit reference to this limit altogether.

2.2 Differentiation of Tensor Fields

We use the term **field** to denote a function which is defined in a region of Euclidean space E^3 . By a **scalar field** we mean a function of the form $\phi : E^3 \rightarrow \mathbb{R}$, by a **vector field** we mean $\mathbf{v} : E^3 \rightarrow \mathcal{V}$ and by a **second-order tensor field** we mean $\mathbf{S} : E^3 \rightarrow \mathcal{V}^2$. In this section we define various differential operations on such tensor fields and discuss some of their properties.

2.2.1 Derivatives, Gradients

Here we define the derivative or gradient of scalar and vector fields. All other differential operations that we perform on tensor fields will be based on these definitions.

Definition 2.3 A scalar field $\phi : \mathbf{E}^3 \rightarrow \mathbf{R}$ is said to be **differentiable** at $\mathbf{x} \in \mathbf{E}^3$ if there exists a vector $\nabla\phi(\mathbf{x}) \in \mathcal{V}$ such that

$$\phi(\mathbf{x} + \mathbf{h}) = \phi(\mathbf{x}) + \nabla\phi(\mathbf{x}) \cdot \mathbf{h} + o(|\mathbf{h}|),$$

or equivalently

$$\nabla\phi(\mathbf{x}) \cdot \mathbf{a} = \left. \frac{d}{d\alpha} \phi(\mathbf{x} + \alpha\mathbf{a}) \right|_{\alpha=0}, \quad \forall \mathbf{a} \in \mathcal{V},$$

where $\alpha \in \mathbf{R}$. The vector $\nabla\phi(\mathbf{x})$ is called the **derivative** or **gradient** of ϕ at $\mathbf{x}$. □

If ϕ is differentiable at $\mathbf{x}$ it can be shown that the vector $\nabla\phi(\mathbf{x})$ is necessarily unique. The second characterization follows from the first upon setting $\mathbf{h} = \alpha\mathbf{a}$, dividing through by α , and taking the limit $\alpha \rightarrow 0$. The following result provides an explicit characterization of the derivative $\nabla\phi(\mathbf{x})$ in any coordinate frame.

Result 2.4 Scalar Gradient in Coordinates. Let $\{\mathbf{e}_i\}$ be an arbitrary frame. Then

$$\nabla\phi(\mathbf{x}) = \frac{\partial\phi}{\partial x_i}(\mathbf{x}) \mathbf{e}_i,$$

where (x_1, x_2, x_3) are the coordinates of $\mathbf{x}$ in $\{\mathbf{e}_i\}$. □

Proof Writing ϕ as a function of the coordinates x_i we have, by a slight abuse of notation, $\phi(\mathbf{x}) = \phi(x_1, x_2, x_3)$. For any scalar α and vector $\mathbf{a} = a_k \mathbf{e}_k$ this gives

$$\phi(\mathbf{x} + \alpha\mathbf{a}) = \phi(x_1 + \alpha a_1, x_2 + \alpha a_2, x_3 + \alpha a_3).$$

By Definition 2.3 and the chain rule we find

$$\begin{aligned} \nabla\phi(\mathbf{x}) \cdot \mathbf{a} &= \left. \frac{d}{d\alpha} \phi(\mathbf{x} + \alpha\mathbf{a}) \right|_{\alpha=0} \\ &= \frac{\partial\phi}{\partial x_1}(\mathbf{x}) a_1 + \frac{\partial\phi}{\partial x_2}(\mathbf{x}) a_2 + \frac{\partial\phi}{\partial x_3}(\mathbf{x}) a_3 \\ &= \frac{\partial\phi}{\partial x_i}(\mathbf{x}) \mathbf{e}_i \cdot a_k \mathbf{e}_k, \end{aligned}$$

which implies $\nabla\phi(\mathbf{x}) = \frac{\partial\phi}{\partial x_i}(\mathbf{x}) \mathbf{e}_i$. □

Remarks:

- (1) The derivative of a scalar field $\phi : \mathbf{E}^3 \rightarrow \mathbf{R}$ is a vector field $\nabla\phi : \mathbf{E}^3 \rightarrow \mathcal{V}$.
- (2) We sometimes denote $\partial\phi/\partial x_i$ by $\phi_{,i}$.
- (3) Since all partial derivatives are assumed to exist and be continuous, we can use Taylor's Theorem to deduce, for any small vector $\mathbf{h} \in \mathcal{V}$

$$\phi(\mathbf{x} + \mathbf{h}) = \phi(\mathbf{x}) + \frac{\partial\phi}{\partial x_i}(\mathbf{x})h_i + \mathcal{O}(|\mathbf{h}|^2),$$

or equivalently

$$\phi(\mathbf{x} + \mathbf{h}) = \phi(\mathbf{x}) + \nabla\phi(\mathbf{x}) \cdot \mathbf{h} + \mathcal{O}(|\mathbf{h}|^2).$$

Thus, in the smooth case, the difference between $\phi(\mathbf{x} + \mathbf{h})$ and $\phi(\mathbf{x}) + \nabla\phi(\mathbf{x}) \cdot \mathbf{h}$ is order $\mathcal{O}(|\mathbf{h}|^2)$, not just $o(|\mathbf{h}|)$ as indicated in the definition of $\nabla\phi(\mathbf{x})$. $\square$

Definition 2.5 A vector field $\mathbf{v} : \mathbf{E}^3 \rightarrow \mathcal{V}$ is said to be **differentiable** at $\mathbf{x} \in \mathbf{E}^3$ if there exists a second-order tensor $\nabla\mathbf{v}(\mathbf{x}) \in \mathcal{V}^2$ such that

$$\mathbf{v}(\mathbf{x} + \mathbf{h}) = \mathbf{v}(\mathbf{x}) + \nabla\mathbf{v}(\mathbf{x})\mathbf{h} + o(|\mathbf{h}|),$$

or equivalently

$$\nabla\mathbf{v}(\mathbf{x})\mathbf{a} = \left. \frac{d}{d\alpha}\mathbf{v}(\mathbf{x} + \alpha\mathbf{a}) \right|_{\alpha=0}, \quad \forall \mathbf{a} \in \mathcal{V},$$

where $\alpha \in \mathbf{R}$. The tensor $\nabla\mathbf{v}(\mathbf{x})$ is called the **derivative** or **gradient** of $\mathbf{v}$ at $\mathbf{x}$. $\square$

If $\mathbf{v}$ is differentiable at $\mathbf{x}$ it can be shown that the tensor $\nabla\mathbf{v}(\mathbf{x})$ is necessarily unique. As before, the second characterization follows from the first upon setting $\mathbf{h} = \alpha\mathbf{a}$, dividing through by α , and taking the limit $\alpha \rightarrow 0$. The following result provides an explicit characterization of the derivative $\nabla\mathbf{v}(\mathbf{x})$ in any coordinate frame.

Result 2.6 Vector Gradient in Coordinates. Let $\{\mathbf{e}_i\}$ be an arbitrary frame and let $\mathbf{v}(\mathbf{x}) = v_i(\mathbf{x})\mathbf{e}_i$. Then

$$\nabla\mathbf{v}(\mathbf{x}) = \frac{\partial v_i}{\partial x_j}(\mathbf{x}) \mathbf{e}_i \otimes \mathbf{e}_j,$$

where (x_1, x_2, x_3) are the coordinates of $\mathbf{x}$ in $\{\mathbf{e}_i\}$. $\square$

Proof Writing the components v_i as functions of the coordinates x_j we have, by a slight abuse of notation, $v_i(\mathbf{x}) = v_i(x_1, x_2, x_3)$. For any scalar α and vector $\mathbf{a} = a_k \mathbf{e}_k$ this gives

$$v_i(\mathbf{x} + \alpha \mathbf{a}) = v_i(x_1 + \alpha a_1, x_2 + \alpha a_2, x_3 + \alpha a_3),$$

and by the chain rule we find

$$\begin{aligned} \left. \frac{d}{d\alpha} v_i(\mathbf{x} + \alpha \mathbf{a}) \right|_{\alpha=0} &= \frac{\partial v_i}{\partial x_1}(\mathbf{x}) a_1 + \frac{\partial v_i}{\partial x_2}(\mathbf{x}) a_2 + \frac{\partial v_i}{\partial x_3}(\mathbf{x}) a_3 \\ &= \frac{\partial v_i}{\partial x_j}(\mathbf{x}) a_j. \end{aligned}$$

Using this result, together with Definition 2.5 and properties of dyadic products (see Chapter 1), we obtain

$$\begin{aligned} \nabla \mathbf{v}(\mathbf{x}) \mathbf{a} &= \left. \frac{d}{d\alpha} \mathbf{v}(\mathbf{x} + \alpha \mathbf{a}) \right|_{\alpha=0} = \left. \frac{d}{d\alpha} v_i(\mathbf{x} + \alpha \mathbf{a}) \right|_{\alpha=0} \mathbf{e}_i \\ &= \frac{\partial v_i}{\partial x_j}(\mathbf{x}) a_j \mathbf{e}_i \\ &= \left(\frac{\partial v_i}{\partial x_j}(\mathbf{x}) \mathbf{e}_i \otimes \mathbf{e}_j \right) a_k \mathbf{e}_k, \end{aligned}$$

which implies $\nabla \mathbf{v}(\mathbf{x}) = \frac{\partial v_i}{\partial x_j}(\mathbf{x}) \mathbf{e}_i \otimes \mathbf{e}_j$. □

Remarks:

- (1) The derivative of a vector field $\mathbf{v} : \mathbb{E}^3 \rightarrow \mathcal{V}$ is a second-order tensor field $\nabla \mathbf{v} : \mathbb{E}^3 \rightarrow \mathcal{V}^2$.
- (2) As for scalar fields, we often denote $\partial v_i / \partial x_j$ by $v_{i,j}$.
- (3) Since all partial derivatives are assumed to exist and be continuous, we can use Taylor's Theorem on each component v_i to deduce, for any small vector $\mathbf{h} \in \mathcal{V}$

$$v_i(\mathbf{x} + \mathbf{h}) = v_i(\mathbf{x}) + \frac{\partial v_i}{\partial x_j}(\mathbf{x}) h_j + \mathcal{O}(|\mathbf{h}|^2),$$

or in tensor notation

$$\mathbf{v}(\mathbf{x} + \mathbf{h}) = \mathbf{v}(\mathbf{x}) + \nabla \mathbf{v}(\mathbf{x}) \mathbf{h} + \mathcal{O}(|\mathbf{h}|^2).$$

Thus, in the smooth case, the difference between $\mathbf{v}(\mathbf{x} + \mathbf{h})$ and $\mathbf{v}(\mathbf{x}) + \nabla \mathbf{v}(\mathbf{x}) \mathbf{h}$ is order $\mathcal{O}(|\mathbf{h}|^2)$, not just $o(|\mathbf{h}|)$ as indicated in the definition of $\nabla \mathbf{v}(\mathbf{x})$.

- (4) Since we identify points $\mathbf{x} \in \mathbf{E}^3$ with their position vectors $\mathbf{x} - \mathbf{o} \in \mathcal{V}$ relative to a fixed origin, any result that we state for vector fields $\mathbf{v} : \mathbf{E}^3 \rightarrow \mathcal{V}$ will also apply to **point mappings**, which are functions of the form $\chi : \mathbf{E}^3 \rightarrow \mathbf{E}^3$. Thus the derivative or gradient of a point mapping χ is a second-order tensor field $\nabla \chi : \mathbf{E}^3 \rightarrow \mathcal{V}^2$, where

$$\nabla \chi(\mathbf{x}) = \frac{\partial \chi_i}{\partial x_j}(\mathbf{x}) \mathbf{e}_i \otimes \mathbf{e}_j.$$

□

2.2.2 Divergence

Here we introduce the divergence of vector and second-order tensor fields. These quantities arise in the statements of various integral theorems introduced later.

Definition 2.7 *To any vector field $\mathbf{v} : \mathbf{E}^3 \rightarrow \mathcal{V}$ we associate a scalar field $\nabla \cdot \mathbf{v} : \mathbf{E}^3 \rightarrow \mathbf{R}$ defined by*

$$\nabla \cdot \mathbf{v} = \text{tr } \nabla \mathbf{v}.$$

*We call $\nabla \cdot \mathbf{v}$ the **divergence** of $\mathbf{v}$.*

□

If we interpret the vector field $\mathbf{v}$ as the velocity field in a flowing fluid or gas, then the scalar $\nabla \cdot \mathbf{v}$ at a point $\mathbf{x}$ can be interpreted as the rate of volume expansion at $\mathbf{x}$. (See the discussion following the Divergence Theorem in Section 2.3.) The next result, which follows directly from Result 2.6 and the definition of the trace function, provides an explicit characterization of the divergence $\nabla \cdot \mathbf{v}$ in any coordinate frame.

Result 2.8 Vector Divergence in Coordinates. *Let $\{\mathbf{e}_i\}$ be an arbitrary frame and let $\mathbf{v}(\mathbf{x}) = v_i(\mathbf{x})\mathbf{e}_i$. Then*

$$(\nabla \cdot \mathbf{v})(\mathbf{x}) = \frac{\partial v_i}{\partial x_i}(\mathbf{x}) = v_{i,i}(\mathbf{x}),$$

where (x_1, x_2, x_3) are the coordinates of $\mathbf{x}$ in $\{\mathbf{e}_i\}$.

□

To formulate balance laws for material bodies it will prove useful to extend the above definition of divergence to second-order tensor fields. While different extensions are possible, we adopt one which leads to a convenient form of the Tensor Divergence Theorem (Result 2.17).

Definition 2.9 To any second-order tensor field $\mathbf{S} : \mathbf{E}^3 \rightarrow \mathcal{V}^2$ we associate a vector field $\nabla \cdot \mathbf{S} : \mathbf{E}^3 \rightarrow \mathcal{V}$ defined by the relation

$$(\nabla \cdot \mathbf{S}) \cdot \mathbf{a} = \nabla \cdot (\mathbf{S}^T \mathbf{a}),$$

for all constant vectors $\mathbf{a}$. We call $\nabla \cdot \mathbf{S}$ the **divergence** of $\mathbf{S}$. □

Thus the divergence of a second-order tensor field is defined using the corresponding notion for vector fields. The following result provides an explicit characterization of $\nabla \cdot \mathbf{S}$ in any coordinate frame.

Result 2.10 Tensor Divergence in Coordinates. Let $\{\mathbf{e}_i\}$ be an arbitrary frame and let $\mathbf{S}(\mathbf{x}) = S_{ij}(\mathbf{x})\mathbf{e}_i \otimes \mathbf{e}_j$. Then

$$(\nabla \cdot \mathbf{S})(\mathbf{x}) = \frac{\partial S_{ij}}{\partial x_j}(\mathbf{x})\mathbf{e}_i = S_{ij,j}(\mathbf{x})\mathbf{e}_i,$$

where (x_1, x_2, x_3) are the coordinates of $\mathbf{x}$ in $\{\mathbf{e}_i\}$. □

Proof Consider an arbitrary, constant vector $\mathbf{a} = a_k \mathbf{e}_k$ and let $\mathbf{q} = \mathbf{S}^T \mathbf{a}$. Then we have $\mathbf{q} = q_j \mathbf{e}_j$, where $q_j = S_{ij}a_i$. Using Definition 2.9 together with Result 2.8 we get

$$(\nabla \cdot \mathbf{S}) \cdot \mathbf{a} = \nabla \cdot \mathbf{q} = q_{j,j} = S_{ij,j}a_i = (S_{ij,j}\mathbf{e}_i) \cdot a_k \mathbf{e}_k,$$

which implies $\nabla \cdot \mathbf{S} = S_{ij,j} \mathbf{e}_i$. □

The next result summarizes some useful product rules for the gradient and divergence operations on scalar, vector and second-order tensor fields. Many other similar rules are established in the exercises.

Result 2.11 Gradient and Divergence Product Rules. Let ϕ , $\mathbf{v}$ and $\mathbf{S}$ be scalar, vector and second-order tensor fields, respectively. Then

$$\begin{aligned} \nabla(\phi \mathbf{v}) &= \mathbf{v} \otimes \nabla \phi + \phi \nabla \mathbf{v}, \\ \nabla \cdot (\phi \mathbf{S}) &= \phi \nabla \cdot \mathbf{S} + \mathbf{S} \nabla \phi, \\ \nabla \cdot (\mathbf{S}^T \mathbf{v}) &= (\nabla \cdot \mathbf{S}) \cdot \mathbf{v} + \mathbf{S} : \nabla \mathbf{v}. \end{aligned}$$

□

Proof Let $\mathbf{v} = v_i \mathbf{e}_i$ and $\mathbf{S} = S_{ij} \mathbf{e}_i \otimes \mathbf{e}_j$. To establish the first result we notice that $\phi \mathbf{v}$ is a vector field and use Result 2.6 to obtain

$$\begin{aligned}\nabla(\phi \mathbf{v}) &= \frac{\partial(\phi v_i)}{\partial x_j} \mathbf{e}_i \otimes \mathbf{e}_j \\ &= (v_i \phi_{,j} + \phi v_{i,j}) \mathbf{e}_i \otimes \mathbf{e}_j \\ &= \mathbf{v} \otimes \nabla \phi + \phi \nabla \mathbf{v}.\end{aligned}$$

To establish the second result we notice that $\phi \mathbf{S}$ is a second-order tensor field and use Result 2.10 to obtain

$$\begin{aligned}\nabla \cdot (\phi \mathbf{S}) &= \frac{\partial(\phi S_{ij})}{\partial x_j} \mathbf{e}_i \\ &= (S_{ij} \phi_{,j} + \phi S_{ij,j}) \mathbf{e}_i \\ &= \mathbf{S} \nabla \phi + \phi \nabla \cdot \mathbf{S}.\end{aligned}$$

To establish the third result we notice that $\mathbf{S}^T \mathbf{v}$ is a vector field, in particular, $\mathbf{S}^T \mathbf{v} = S_{ij} v_i \mathbf{e}_j$. Using Result 2.8 we obtain

$$\begin{aligned}\nabla \cdot (\mathbf{S}^T \mathbf{v}) &= \frac{\partial(S_{ij} v_i)}{\partial x_j} \\ &= S_{ij,j} v_i + S_{ij} v_{i,j} \\ &= (\nabla \cdot \mathbf{S}) \cdot \mathbf{v} + \mathbf{S} : \nabla \mathbf{v}.\end{aligned}$$

□

2.2.3 Curl

Here we introduce the curl of a vector field. This quantity arises in the statement of Stokes' Theorem introduced later and plays an important role in describing the motions of fluids and gases.

Definition 2.12 *To any vector field $\mathbf{v} : \mathbf{E}^3 \rightarrow \mathcal{V}$ we associate another vector field $\nabla \times \mathbf{v} : \mathbf{E}^3 \rightarrow \mathcal{V}$ defined by*

$$(\nabla \times \mathbf{v}) \times \mathbf{a} = (\nabla \mathbf{v} - \nabla \mathbf{v}^T) \mathbf{a},$$

*for all constant vectors $\mathbf{a}$. We call $\nabla \times \mathbf{v}$ the **curl** of $\mathbf{v}$.*

□

If we interpret the vector field $\mathbf{v}$ as the velocity field in a flowing fluid or gas, then the vector $\nabla \times \mathbf{v}$ at a point $\mathbf{x}$ provides information on the rate and direction of rotation at $\mathbf{x}$. (See the discussion following Stokes' Theorem in Section 2.3.) The following result provides an explicit characterization of the curl $\nabla \times \mathbf{v}$ in any coordinate frame.

Result 2.13 Vector Curl in Coordinates. Let $\{\mathbf{e}_i\}$ be an arbitrary frame and let $\mathbf{v}(\mathbf{x}) = v_i(\mathbf{x})\mathbf{e}_i$. Then

$$(\nabla \times \mathbf{v})(\mathbf{x}) = \epsilon_{ijk} v_{i,k}(\mathbf{x}) \mathbf{e}_j,$$

where (x_1, x_2, x_3) are the coordinates of $\mathbf{x}$ in $\{\mathbf{e}_i\}$. Equivalently, we have

$$\nabla \times \mathbf{v} = \left(\frac{\partial v_3}{\partial x_2} - \frac{\partial v_2}{\partial x_3} \right) \mathbf{e}_1 + \left(\frac{\partial v_1}{\partial x_3} - \frac{\partial v_3}{\partial x_1} \right) \mathbf{e}_2 + \left(\frac{\partial v_2}{\partial x_1} - \frac{\partial v_1}{\partial x_2} \right) \mathbf{e}_3.$$

□

Proof In view of Result 1.3 we notice that $\nabla \times \mathbf{v}$ is the axial vector associated with the skew-symmetric tensor $\nabla \mathbf{v} - \nabla \mathbf{v}^T$. If we let $\mathbf{w} = \nabla \times \mathbf{v}$ and $\mathbf{T} = \nabla \mathbf{v} - \nabla \mathbf{v}^T$, then in any frame we have, by Result 1.3

$$w_j = \frac{1}{2} \epsilon_{ijk} T_{ik} = \frac{1}{2} \epsilon_{ijk} (v_{i,k} - v_{k,i}).$$

Carrying out the multiplication on the right-hand side of the last equation and using the fact that $\epsilon_{ijk} = -\epsilon_{kji}$ we get

$$w_j = \frac{1}{2} (\epsilon_{ijk} v_{i,k} - \epsilon_{ijk} v_{k,i}) = \frac{1}{2} (\epsilon_{ijk} v_{i,k} + \epsilon_{kji} v_{k,i}) = \epsilon_{ijk} v_{i,k},$$

where the last equality follows from the fact that i and k are dummy indices. Since $\mathbf{w} = \nabla \times \mathbf{v}$ we obtain

$$\nabla \times \mathbf{v} = \epsilon_{ijk} v_{i,k} \mathbf{e}_j.$$

The second result is obtained by carrying out the indicated sums using the definition of the permutation symbol. □

2.2.4 Laplacian

Here we introduce the Laplacian of scalar and vector fields. These quantities are second-order differential operations that arise frequently in applications.

Definition 2.14 To any scalar field $\phi : \mathbf{E}^3 \rightarrow \mathbf{R}$ we associate another scalar field $\Delta \phi : \mathbf{E}^3 \rightarrow \mathbf{R}$ defined by

$$\Delta \phi = \nabla \cdot (\nabla \phi),$$

and to any vector field $\mathbf{v} : \mathbf{E}^3 \rightarrow \mathcal{V}$ we associate another vector field $\Delta \mathbf{v} : \mathbf{E}^3 \rightarrow \mathcal{V}$ defined by

$$\Delta \mathbf{v} = \nabla \cdot (\nabla \mathbf{v}).$$

We call $\Delta\phi$ and $\Delta\mathbf{v}$ the **Laplacians** of ϕ and $\mathbf{v}$, respectively. $\square$

Thus the Laplacian of a scalar or vector field is the divergence of the corresponding gradient. The following result provides an explicit characterization of $\Delta\phi$ and $\Delta\mathbf{v}$ in any coordinate frame.

Result 2.15 Scalar and Vector Laplacian in Coordinates. *Let $\{\mathbf{e}_i\}$ be an arbitrary frame and let $\mathbf{v}(\mathbf{x}) = v_i(\mathbf{x})\mathbf{e}_i$. Then*

$$\Delta\phi(\mathbf{x}) = \phi_{,ii}(\mathbf{x}) \quad \text{and} \quad \Delta\mathbf{v}(\mathbf{x}) = v_{i,jj}(\mathbf{x})\mathbf{e}_i,$$

where (x_1, x_2, x_3) are the coordinates of $\mathbf{x}$ in $\{\mathbf{e}_i\}$. $\square$

Proof By Result 2.4 we have $\nabla\phi = \phi_{,i}\mathbf{e}_i$ and using Result 2.8 we obtain

$$\Delta\phi = \nabla \cdot (\nabla\phi) = (\phi_{,i})_{,i} = \phi_{,ii}.$$

Similarly, by Result 2.6 we have $\nabla\mathbf{v} = v_{i,j}\mathbf{e}_i \otimes \mathbf{e}_j$ and using Result 2.10 we obtain

$$\Delta\mathbf{v} = \nabla \cdot (\nabla\mathbf{v}) = (v_{i,j})_{,j}\mathbf{e}_i = v_{i,jj}\mathbf{e}_i.$$

$\square$

2.3 Integral Theorems

In this section we state some integral theorems that will be useful later when we derive local statements of balance laws for continuum bodies.

2.3.1 Divergence Theorem

Here we state, without proof, a result that provides a relationship between the volume integral of the divergence of a vector field in a region B of $\mathbb{E}^3$, and the surface integral of an associated field over the bounding surface ∂B of B ; we then generalize the result to second-order tensor fields. The validity of these results requires certain assumptions on the region B . For our purposes it will be sufficient to only consider those regions that are **regular** in the following sense: (i) B consists of a finite number of open, disjoint and bounded components; (ii) the bounding surface ∂B is piecewise smooth and consists of a finite number of disjoint components; (iii) each component of ∂B is orientable in the sense

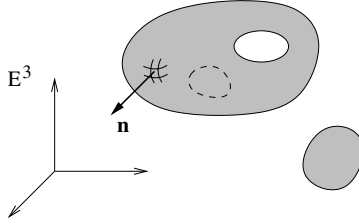


Fig. 2.1 Graphical illustration of a regular region consisting of two disjoint components. One component has an internal void and a hole.

that it clearly has two sides. The notion of a regular region of $\mathbf{E}^3$ encompasses many bodies of interest in mechanics, including those with a finite number of holes and voids as depicted in Figure 2.1.

In the following, dV denotes an infinitesimal volume element in B and dA denotes an infinitesimal area element in ∂B .

Result 2.16 Divergence Theorem. *Let B denote a regular region in $\mathbf{E}^3$ with piecewise smooth boundary ∂B , and consider an arbitrary vector field $\mathbf{v} : B \rightarrow \mathcal{V}$. Then*

$$\int_{\partial B} \mathbf{v} \cdot \mathbf{n} \, dA = \int_B \nabla \cdot \mathbf{v} \, dV,$$

or in components

$$\int_{\partial B} v_i n_i \, dA = \int_B v_{i,i} \, dV,$$

where $\mathbf{n}$ is the outward unit normal field on ∂B . The quantity appearing on the left-hand side of the above relation is called the **flux** of $\mathbf{v}$ across the oriented surface ∂B . □

The Divergence Theorem can be used to gain some insight into the physical meaning of the divergence of a vector field. To this end, consider an arbitrary point $\mathbf{y} \in \mathbf{E}^3$ and let Ω_δ denote a ball centered at $\mathbf{y}$ with radius δ and boundary $\partial\Omega_\delta$. Then by the Divergence Theorem we have

$$\begin{aligned} \int_{\partial\Omega_\delta} \mathbf{v}(\mathbf{x}) \cdot \mathbf{n}(\mathbf{x}) \, dA &= \int_{\Omega_\delta} (\nabla \cdot \mathbf{v})(\mathbf{x}) \, dV \\ &= \int_{\Omega_\delta} [(\nabla \cdot \mathbf{v})(\mathbf{y}) + \mathcal{O}(\delta)] \, dV \\ &= \text{vol}(\Omega_\delta) [(\nabla \cdot \mathbf{v})(\mathbf{y}) + \mathcal{O}(\delta)], \end{aligned}$$

which implies

$$(\nabla \cdot \mathbf{v})(\mathbf{y}) = \lim_{\delta \rightarrow 0} \frac{1}{\text{vol}(\Omega_\delta)} \int_{\partial\Omega_\delta} \mathbf{v} \cdot \mathbf{n} \, dA.$$

Thus, for small δ , we see that $(\nabla \cdot \mathbf{v})(\mathbf{y})$ is the flux of $\mathbf{v}$ across $\partial\Omega_\delta$ divided by the volume of Ω_δ . If we interpret $\mathbf{v}$ as the velocity field in a fluid, then the flux of $\mathbf{v}$ across $\partial\Omega_\delta$ is equal to the net rate (volume per unit time) at which fluid is exiting Ω_δ . In this sense $(\nabla \cdot \mathbf{v})(\mathbf{y})$ provides a measure of volume expansion at $\mathbf{y}$.

The following generalization of the Divergence Theorem to second-order tensor fields will be useful.

Result 2.17 Tensor Divergence Theorem. *Let B denote a regular region in $\mathbb{E}^3$ with piecewise smooth boundary ∂B , and consider an arbitrary second-order tensor field $\mathbf{S} : B \rightarrow \mathcal{V}^2$. Then*

$$\int_{\partial B} \mathbf{S} \mathbf{n} \, dA = \int_B \nabla \cdot \mathbf{S} \, dV,$$

or in components

$$\int_{\partial B} S_{ij} n_j \, dA = \int_B S_{ij,j} \, dV,$$

where $\mathbf{n}$ is the outward unit normal field on ∂B . □

Proof Let $\mathbf{a}$ be an arbitrary, constant vector. Then

$$\mathbf{a} \cdot \int_{\partial B} \mathbf{S} \mathbf{n} \, dA = \int_{\partial B} \mathbf{a} \cdot \mathbf{S} \mathbf{n} \, dA = \int_{\partial B} (\mathbf{S}^T \mathbf{a}) \cdot \mathbf{n} \, dA.$$

Since $\mathbf{S}^T \mathbf{a}$ is a vector field we can apply the Divergence Theorem in Result 2.16 to get

$$\int_{\partial B} (\mathbf{S}^T \mathbf{a}) \cdot \mathbf{n} \, dA = \int_B \nabla \cdot (\mathbf{S}^T \mathbf{a}) \, dV,$$

and by Definition 2.9 we obtain

$$\int_B \nabla \cdot (\mathbf{S}^T \mathbf{a}) \, dV = \int_B \mathbf{a} \cdot (\nabla \cdot \mathbf{S}) \, dV = \mathbf{a} \cdot \int_B \nabla \cdot \mathbf{S} \, dV.$$

Thus from the arbitrariness of $\mathbf{a}$ we deduce

$$\int_{\partial B} \mathbf{S} \mathbf{n} \, dA = \int_B \nabla \cdot \mathbf{S} \, dV.$$

□

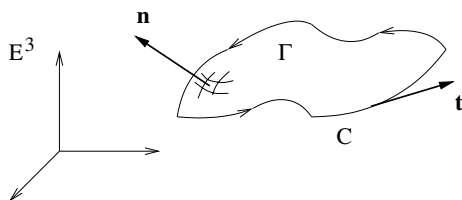


Fig. 2.2 Graphical illustration of a surface Γ with a bounding curve C .

2.3.2 Stokes' Theorem

Here we state, without proof, a result that provides a relationship between the line integral of a vector field along a simple closed curve C and the surface integral of an associated field over a surface Γ whose boundary is $C = \partial\Gamma$ as shown in Figure 2.2. The validity of this result requires certain assumptions on C and Γ . For our purposes it will be sufficient to only consider curves C which do not intersect themselves, and which are bounded and piecewise smooth. Moreover, we assume that the surface Γ is orientable, bounded and piecewise smooth. The orientations for the unit tangent field $\mathbf{t}$ on C and the unit normal field $\mathbf{n}$ on Γ are assumed to be chosen as follows: a person walking around C in the direction $\mathbf{t}$ with their head in the direction $\mathbf{n}$ has the surface Γ to their left.

In the following, dA is an infinitesimal area element in Γ and ds is an infinitesimal length element in C .

Result 2.18 Stokes' Theorem. *Let Γ be a surface in $\mathbb{E}^3$ with a piecewise smooth boundary curve $C = \partial\Gamma$, and consider an arbitrary vector field $\mathbf{v} : \mathbb{E}^3 \rightarrow \mathcal{V}$. Then*

$$\int_{\Gamma} (\nabla \times \mathbf{v}) \cdot \mathbf{n} \, dA = \int_C \mathbf{v} \cdot \mathbf{t} \, ds,$$

where $\mathbf{n}$ is a unit normal field on Γ and $\mathbf{t}$ is a unit tangent field on C oriented as discussed above. The quantity appearing on the right-hand side of the above relation is called the **circulation** of $\mathbf{v}$ around the oriented curve C . □

Stokes' Theorem can be used to gain some insight into the physical meaning of the curl of a vector field. To this end, consider an arbitrary point $\mathbf{y} \in \mathbb{E}^3$ and let Γ_{δ} be a disc centered at $\mathbf{y}$ with radius δ and

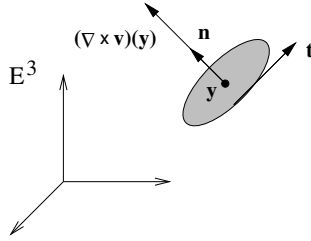


Fig. 2.3 Interpretation of the curl of a vector field in terms of circulation.

boundary $\partial\Gamma_\delta$. Then by Stokes' Theorem we have

$$\begin{aligned} \int_{\partial\Gamma_\delta} \mathbf{v}(\mathbf{x}) \cdot \mathbf{t}(\mathbf{x}) \, ds &= \int_{\Gamma_\delta} (\nabla \times \mathbf{v})(\mathbf{x}) \cdot \mathbf{n}(\mathbf{x}) \, dA \\ &= \int_{\Gamma_\delta} [(\nabla \times \mathbf{v})(\mathbf{y}) \cdot \mathbf{n}(\mathbf{y}) + \mathcal{O}(\delta)] \, dA \\ &= \text{area}(\Gamma_\delta) [(\nabla \times \mathbf{v})(\mathbf{y}) \cdot \mathbf{n}(\mathbf{y}) + \mathcal{O}(\delta)], \end{aligned}$$

which implies

$$\mathbf{n}(\mathbf{y}) \cdot (\nabla \times \mathbf{v})(\mathbf{y}) = \lim_{\delta \rightarrow 0} \frac{1}{\text{area}(\Gamma_\delta)} \int_{\partial\Gamma_\delta} \mathbf{v} \cdot \mathbf{t} \, ds.$$

At the point $\mathbf{y}$ in question suppose $(\nabla \times \mathbf{v})(\mathbf{y}) \neq \mathbf{0}$ and choose the disc Γ_δ such that its normal is given by $\mathbf{n}(\mathbf{y}) = (\nabla \times \mathbf{v})(\mathbf{y}) / |(\nabla \times \mathbf{v})(\mathbf{y})|$ as shown in Figure 2.3. For this choice of Γ_δ we obtain

$$|(\nabla \times \mathbf{v})(\mathbf{y})| = \lim_{\delta \rightarrow 0} \frac{1}{\text{area}(\Gamma_\delta)} \int_{\partial\Gamma_\delta} \mathbf{v} \cdot \mathbf{t} \, ds.$$

Thus, for small δ , we see that $|(\nabla \times \mathbf{v})(\mathbf{y})|$ is the circulation of $\mathbf{v}$ around $\partial\Gamma_\delta$ divided by the area of Γ_δ . Here Γ_δ is a disc of radius δ centered at $\mathbf{y}$ with normal $\mathbf{n}$ in the direction of $(\nabla \times \mathbf{v})(\mathbf{y})$. If we interpret $\mathbf{v}$ as the velocity field in a fluid, then $(\nabla \times \mathbf{v})(\mathbf{y})$ can be interpreted as twice the local angular velocity of the fluid at $\mathbf{y}$.

2.3.3 Localization Theorem

Here we establish a result which states that, if the integral of a function over arbitrary subsets of its domain is zero, then the function itself must be zero. This result will be used frequently in later chapters.

Result 2.19 Localization Theorem. Consider an open set B in $\mathbf{E}^3$ and a continuous function $\phi : B \rightarrow \mathbf{R}$, and let Ω denote an arbitrary open subset of B . If

$$\int_{\Omega} \phi(\mathbf{x}) \, dV = 0, \quad \forall \Omega \subseteq B, \quad (2.1)$$

then

$$\phi(\mathbf{x}) = 0, \quad \forall \mathbf{x} \in B. \quad (2.2)$$

□

Proof The result follows by contradiction. To begin, assume the hypothesis (2.1) is true and assume the statement (2.2) is false; that is, there exists some point $\mathbf{y} \in B$ for which $\phi(\mathbf{y}) \neq 0$, say $\phi(\mathbf{y}) = 2\delta > 0$; the case $\delta < 0$ is dealt with similarly. Then, by continuity of ϕ and the fact that B is open, there exists an open neighborhood Ω of $\mathbf{y}$ such that $\Omega \subset B$ and $\phi(\mathbf{x}) > \delta$ for all $\mathbf{x} \in \Omega$. For this subset we have

$$\int_{\Omega} \phi(\mathbf{x}) \, dV > \delta \, \text{vol}[\Omega] > 0,$$

which contradicts (2.1). Hence the statement (2.2) must be true. □

Remarks:

- (1) Result 2.19 also holds when the scalar field ϕ is replaced by a vector field $\mathbf{v}$ or second-order tensor field $\mathbf{S}$. In particular, the result for scalar fields can be applied to each component of a vector or second-order tensor field.
- (2) Result 2.19 can also be generalized to integrals other than volume integrals. For example, if Γ is an open disc and $\int_{\Omega} \phi(\mathbf{x}) \, dA = 0$ for all open subsets $\Omega \subseteq \Gamma$, then a similar argument shows that $\phi(\mathbf{x}) = 0$ for all $\mathbf{x} \in \Gamma$. □

2.4 Functions of Second-Order Tensors

Here we define the derivative of functions of the form $\psi : \mathcal{V}^2 \rightarrow \mathbf{R}$, such as the determinant and trace functions, and functions of the form $\Sigma : \mathcal{V}^2 \rightarrow \mathcal{V}^2$, such as the exponential function and the stress response functions to be considered in Chapters 7 and 9.

2.4.1 Scalar-Valued Functions

Definition 2.20 A function $\psi : \mathcal{V}^2 \rightarrow \mathcal{R}$ is said to be **differentiable** at $\mathbf{A} \in \mathcal{V}^2$ if there exists a second-order tensor $D\psi(\mathbf{A}) \in \mathcal{V}^2$ such that

$$\psi(\mathbf{A} + \mathbf{H}) = \psi(\mathbf{A}) + D\psi(\mathbf{A}) : \mathbf{H} + o(|\mathbf{H}|),$$

or equivalently

$$D\psi(\mathbf{A}) : \mathbf{B} = \left. \frac{d}{d\alpha} \psi(\mathbf{A} + \alpha \mathbf{B}) \right|_{\alpha=0}, \quad \forall \mathbf{B} \in \mathcal{V}^2,$$

where $\alpha \in \mathcal{R}$. The tensor $D\psi(\mathbf{A})$ is called the **derivative** of ψ at $\mathbf{A}$. $\square$

If ψ is differentiable at $\mathbf{A}$ it can be shown that the tensor $D\psi(\mathbf{A})$ is necessarily unique. The second characterization follows from the first upon setting $\mathbf{H} = \alpha \mathbf{B}$, dividing through by α , and taking the limit $\alpha \rightarrow 0$. The following result provides an explicit characterization of the derivative $D\psi(\mathbf{A})$ in any coordinate frame.

Result 2.21 Derivative of Scalar Function in Components. Let $\{\mathbf{e}_i\}$ be an arbitrary frame. Then

$$D\psi(\mathbf{A}) = \frac{\partial \psi}{\partial A_{ij}}(\mathbf{A}) \mathbf{e}_i \otimes \mathbf{e}_j,$$

where $(A_{11}, A_{12}, \dots, A_{33})$ are the components of $\mathbf{A}$ in the frame $\{\mathbf{e}_i\}$. $\square$

Proof Writing ψ as a function of the components A_{ij} we have, by a slight abuse of notation, $\psi(\mathbf{A}) = \psi(A_{11}, A_{12}, \dots, A_{33})$. For any scalar α and tensor $\mathbf{B} = B_{kl} \mathbf{e}_k \otimes \mathbf{e}_l$ this gives

$$\psi(\mathbf{A} + \alpha \mathbf{B}) = \psi(A_{11} + \alpha B_{11}, \dots, A_{33} + \alpha B_{33}).$$

By Definition 2.20 and the chain rule we find

$$\begin{aligned} D\psi(\mathbf{A}) : \mathbf{B} &= \left. \frac{d}{d\alpha} \psi(\mathbf{A} + \alpha \mathbf{B}) \right|_{\alpha=0} \\ &= \frac{\partial \psi}{\partial A_{11}}(\mathbf{A}) B_{11} + \dots + \frac{\partial \psi}{\partial A_{33}}(\mathbf{A}) B_{33} \\ &= \left(\frac{\partial \psi}{\partial A_{ij}}(\mathbf{A}) \mathbf{e}_i \otimes \mathbf{e}_j \right) : B_{kl} \mathbf{e}_k \otimes \mathbf{e}_l, \end{aligned}$$

which implies $D\psi(\mathbf{A}) = \frac{\partial \psi}{\partial A_{ij}}(\mathbf{A}) \mathbf{e}_i \otimes \mathbf{e}_j$. $\square$

Remarks:

- (1) The derivative of a scalar-valued function $\psi : \mathcal{V}^2 \rightarrow \mathbb{R}$ is a second-order tensor-valued function $D\psi : \mathcal{V}^2 \rightarrow \mathcal{V}^2$.
- (2) Since all partial derivatives are assumed to exist and be continuous, we can use Taylor's Theorem to deduce, for any small tensor $\mathbf{H} \in \mathcal{V}^2$

$$\psi(\mathbf{A} + \mathbf{H}) = \psi(\mathbf{A}) + \frac{\partial \psi}{\partial A_{ij}}(\mathbf{A}) H_{ij} + \mathcal{O}(|\mathbf{H}|^2),$$

or equivalently

$$\psi(\mathbf{A} + \mathbf{H}) = \psi(\mathbf{A}) + D\psi(\mathbf{A}) : \mathbf{H} + \mathcal{O}(|\mathbf{H}|^2).$$

Thus, in the smooth case, the difference between $\psi(\mathbf{A} + \mathbf{H})$ and $\psi(\mathbf{A}) + D\psi(\mathbf{A}) : \mathbf{H}$ is order $\mathcal{O}(|\mathbf{H}|^2)$, not just $o(|\mathbf{H}|)$ as indicated in the definition of $D\psi(\mathbf{A})$. $\square$

Example: Consider the function $\psi : \mathcal{V}^2 \rightarrow \mathbb{R}$ defined by

$$\psi(\mathbf{A}) = \frac{1}{2} \mathbf{A} : \mathbf{A} = \frac{1}{2} A_{kl} A_{kl}.$$

To compute $D\psi(\mathbf{A})$ we first take the partial derivative of ψ with respect to a general component A_{ij} to obtain

$$\frac{\partial \psi}{\partial A_{ij}}(\mathbf{A}) = \frac{1}{2} \frac{\partial A_{kl}}{\partial A_{ij}} A_{kl} + \frac{1}{2} A_{kl} \frac{\partial A_{kl}}{\partial A_{ij}} = \frac{\partial A_{kl}}{\partial A_{ij}} A_{kl}.$$

Since $\partial A_{kl} / \partial A_{ij} = \delta_{ki} \delta_{lj}$ we find

$$\frac{\partial \psi}{\partial A_{ij}}(\mathbf{A}) = \delta_{ki} \delta_{lj} A_{kl} = A_{ij},$$

from which we deduce $D\psi(\mathbf{A}) = A_{ij} \mathbf{e}_i \otimes \mathbf{e}_j = \mathbf{A}$. $\square$

Result 2.22 Derivative of Determinant. Let $\psi(\mathbf{S}) = \det \mathbf{S}$. If $\mathbf{A}$ is invertible, then the derivative of ψ at $\mathbf{A}$ is $D\psi(\mathbf{A}) = \det(\mathbf{A}) \mathbf{A}^{-T}$. $\square$

Proof Let $\mathbf{B} \in \mathcal{V}^2$ be arbitrary. Then

$$\begin{aligned} \psi(\mathbf{A} + \alpha \mathbf{B}) &= \det(\mathbf{A} + \alpha \mathbf{B}) \\ &= \det(\alpha \mathbf{A}(\mathbf{A}^{-1} \mathbf{B} - \lambda \mathbf{I})) \\ &= \det(\alpha \mathbf{A}) \det(\mathbf{A}^{-1} \mathbf{B} - \lambda \mathbf{I}), \end{aligned}$$

where $\lambda = -1/\alpha$. By definition of the determinant of $\alpha \mathbf{A} \in \mathcal{V}^2$ we have

$$\det(\alpha \mathbf{A}) = \alpha^3 \det \mathbf{A},$$

and by definition of the principal invariants of the second-order tensor $\mathbf{A}^{-1} \mathbf{B}$ we find

$$\begin{aligned} \det(\mathbf{A}^{-1} \mathbf{B} - \lambda \mathbf{I}) &= -\lambda^3 + \lambda^2 I_1(\mathbf{A}^{-1} \mathbf{B}) - \lambda I_2(\mathbf{A}^{-1} \mathbf{B}) + I_3(\mathbf{A}^{-1} \mathbf{B}) \\ &= \frac{1}{\alpha^3} + \frac{1}{\alpha^2} I_1(\mathbf{A}^{-1} \mathbf{B}) + \frac{1}{\alpha} I_2(\mathbf{A}^{-1} \mathbf{B}) + I_3(\mathbf{A}^{-1} \mathbf{B}), \end{aligned}$$

which implies

$$\begin{aligned} \psi(\mathbf{A} + \alpha \mathbf{B}) &= \det(\mathbf{A}) + \alpha \det(\mathbf{A}) I_1(\mathbf{A}^{-1} \mathbf{B}) \\ &\quad + \alpha^2 \det(\mathbf{A}) I_2(\mathbf{A}^{-1} \mathbf{B}) + \alpha^3 \det(\mathbf{A}) I_3(\mathbf{A}^{-1} \mathbf{B}). \end{aligned}$$

From this expression and Definition 2.20 we deduce

$$D\psi(\mathbf{A}) : \mathbf{B} = \det(\mathbf{A}) I_1(\mathbf{A}^{-1} \mathbf{B}),$$

and since $I_1(\mathbf{A}^{-1} \mathbf{B}) = \text{tr}(\mathbf{A}^{-1} \mathbf{B}) = \mathbf{A}^{-T} : \mathbf{B}$ we find

$$D\psi(\mathbf{A}) : \mathbf{B} = \det(\mathbf{A}) \mathbf{A}^{-T} : \mathbf{B}.$$

The desired result $D\psi(\mathbf{A}) = \det(\mathbf{A}) \mathbf{A}^{-T}$ follows from the arbitrariness of $\mathbf{B}$. $\square$

Result 2.23 Time Derivative of Determinant. *Let $\mathbf{S}$ be a time-dependent second-order tensor, that is, $\mathbf{S} : \mathbf{R} \rightarrow \mathcal{V}^2$. Then*

$$\frac{d}{dt}(\det \mathbf{S}) = (\det \mathbf{S}) \text{tr}(\mathbf{S}^{-1} \dot{\mathbf{S}}) = \det(\mathbf{S}) \mathbf{S}^{-T} : \dot{\mathbf{S}},$$

where

$$\dot{\mathbf{S}} = \frac{d\mathbf{S}}{dt} = \frac{dS_{ij}}{dt} \mathbf{e}_i \otimes \mathbf{e}_j.$$

$\square$

Proof Let $\psi(\mathbf{S}) = \det \mathbf{S}$. Then

$$\frac{d}{dt} \psi(\mathbf{S}) = \frac{d}{dt} \psi(S_{11}, S_{12}, \dots, S_{33}),$$

and by the chain rule we have

$$\frac{d}{dt} \psi(\mathbf{S}) = \frac{\partial \psi}{\partial S_{11}}(\mathbf{S}) \frac{dS_{11}}{dt} + \dots + \frac{\partial \psi}{\partial S_{33}}(\mathbf{S}) \frac{dS_{33}}{dt} = D\psi(\mathbf{S}) : \dot{\mathbf{S}}.$$

Since $D\psi(\mathbf{S}) = \det(\mathbf{S})\mathbf{S}^{-T}$ we obtain

$$\frac{d}{dt}\psi(\mathbf{S}) = \det(\mathbf{S})\mathbf{S}^{-T} : \dot{\mathbf{S}} = \det(\mathbf{S}) \operatorname{tr}(\mathbf{S}^{-1}\dot{\mathbf{S}}).$$

□

2.4.2 Tensor-Valued Functions

Definition 2.24 A function $\Sigma : \mathcal{V}^2 \rightarrow \mathcal{V}^2$ is said to be **differentiable** at $\mathbf{A} \in \mathcal{V}^2$ if there exists a fourth-order tensor $D\Sigma(\mathbf{A}) \in \mathcal{V}^4$ such that

$$\Sigma(\mathbf{A} + \mathbf{H}) = \Sigma(\mathbf{A}) + D\Sigma(\mathbf{A})\mathbf{H} + o(|\mathbf{H}|),$$

or equivalently

$$D\Sigma(\mathbf{A})\mathbf{B} = \left. \frac{d}{d\alpha} \Sigma(\mathbf{A} + \alpha\mathbf{B}) \right|_{\alpha=0}, \quad \forall \mathbf{B} \in \mathcal{V}^2,$$

where $\alpha \in \mathbb{R}$. The tensor $D\Sigma(\mathbf{A})$ is called the **derivative** of Σ at $\mathbf{A}$.

□

If Σ is differentiable at $\mathbf{A}$ it can be shown that the tensor $D\Sigma(\mathbf{A})$ is necessarily unique. The second characterization follows from the first upon setting $\mathbf{H} = \alpha\mathbf{B}$, dividing through by α , and taking the limit $\alpha \rightarrow 0$. The following result provides an explicit characterization of the derivative $D\Sigma(\mathbf{A})$ in any coordinate frame.

Result 2.25 Derivative of Tensor Function in Components. Let $\{\mathbf{e}_i\}$ be an arbitrary frame and let $\Sigma(\mathbf{A}) = \Sigma_{ij}(\mathbf{A})\mathbf{e}_i \otimes \mathbf{e}_j$. Then

$$D\Sigma(\mathbf{A}) = \frac{\partial \Sigma_{ij}}{\partial A_{kl}}(\mathbf{A})\mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l,$$

where $(A_{11}, A_{12}, \dots, A_{33})$ are the components of $\mathbf{A}$ in the frame $\{\mathbf{e}_i\}$. □

Proof Writing the components Σ_{ij} as functions of the components A_{kl} we have, by a slight abuse of notation, $\Sigma_{ij}(\mathbf{A}) = \Sigma_{ij}(A_{11}, A_{12}, \dots, A_{33})$. For any scalar α and tensor $\mathbf{B} = B_{kl}\mathbf{e}_k \otimes \mathbf{e}_l$ this gives

$$\Sigma_{ij}(\mathbf{A} + \alpha\mathbf{B}) = \Sigma_{ij}(A_{11} + \alpha B_{11}, \dots, A_{33} + \alpha B_{33}),$$

and by the chain rule we find

$$\begin{aligned} \left. \frac{d}{d\alpha} \Sigma_{ij}(\mathbf{A} + \alpha \mathbf{B}) \right|_{\alpha=0} &= \frac{\partial \Sigma_{ij}}{\partial A_{11}}(\mathbf{A}) B_{11} + \cdots + \frac{\partial \Sigma_{ij}}{\partial A_{33}}(\mathbf{A}) B_{33} \\ &= \frac{\partial \Sigma_{ij}}{\partial A_{kl}}(\mathbf{A}) B_{kl}. \end{aligned}$$

Using this result, together with Definition 2.24 and properties of dyadic products (see Chapter 1), we obtain

$$\begin{aligned} D\Sigma(\mathbf{A})\mathbf{B} &= \left. \frac{d}{d\alpha} \Sigma(\mathbf{A} + \alpha \mathbf{B}) \right|_{\alpha=0} \\ &= \left. \frac{d}{d\alpha} \Sigma_{ij}(\mathbf{A} + \alpha \mathbf{B}) \right|_{\alpha=0} \mathbf{e}_i \otimes \mathbf{e}_j \\ &= \frac{\partial \Sigma_{ij}}{\partial A_{kl}}(\mathbf{A}) B_{kl} \mathbf{e}_i \otimes \mathbf{e}_j \\ &= \left(\frac{\partial \Sigma_{ij}}{\partial A_{kl}}(\mathbf{A}) \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l \right) \mathbf{B}, \end{aligned}$$

which implies $D\Sigma(\mathbf{A}) = \frac{\partial \Sigma_{ij}}{\partial A_{kl}}(\mathbf{A}) \mathbf{e}_i \otimes \mathbf{e}_j \otimes \mathbf{e}_k \otimes \mathbf{e}_l$. □

Remarks:

- (1) The derivative of a second-order tensor-valued function $\Sigma : \mathcal{V}^2 \rightarrow \mathcal{V}^2$ is a fourth-order tensor-valued function $D\Sigma : \mathcal{V}^2 \rightarrow \mathcal{V}^4$.
- (2) Since all partial derivatives are assumed to exist and be continuous, we can use Taylor's Theorem to deduce, for any small tensor $\mathbf{H} \in \mathcal{V}^2$

$$\Sigma_{ij}(\mathbf{A} + \mathbf{H}) = \Sigma_{ij}(\mathbf{A}) + \frac{\partial \Sigma_{ij}}{\partial A_{kl}}(\mathbf{A}) H_{kl} + \mathcal{O}(|\mathbf{H}|^2),$$

or in tensor notation

$$\Sigma(\mathbf{A} + \mathbf{H}) = \Sigma(\mathbf{A}) + D\Sigma(\mathbf{A})\mathbf{H} + \mathcal{O}(|\mathbf{H}|^2).$$

Thus, in the smooth case, the difference between $\Sigma(\mathbf{A} + \mathbf{H})$ and $\Sigma(\mathbf{A}) + D\Sigma(\mathbf{A})\mathbf{H}$ is order $\mathcal{O}(|\mathbf{H}|^2)$, not just $o(|\mathbf{H}|)$ as indicated in the definition of $D\Sigma(\mathbf{A})$. □

Example: Consider the function $\Sigma : \mathcal{V}^2 \rightarrow \mathcal{V}^2$ defined by

$$\Sigma(\mathbf{A}) = \text{tr}(\mathbf{A})\mathbf{A} \quad \text{or in components} \quad \Sigma_{ij}(\mathbf{A}) = A_{mm} A_{ij}.$$

The components $[D\boldsymbol{\Sigma}(\mathbf{A})]_{ijkl}$ of the derivative $D\boldsymbol{\Sigma}(\mathbf{A})$ are

$$\begin{aligned} [D\boldsymbol{\Sigma}(\mathbf{A})]_{ijkl} &= \frac{\partial \Sigma_{ij}}{\partial A_{kl}}(\mathbf{A}) \\ &= \frac{\partial A_{mm}}{\partial A_{kl}} A_{ij} + A_{mm} \frac{\partial A_{ij}}{\partial A_{kl}} \\ &= \delta_{mk} \delta_{ml} A_{ij} + A_{mm} \delta_{ik} \delta_{jl} \\ &= \delta_{kl} A_{ij} + A_{mm} \delta_{ik} \delta_{jl}. \end{aligned}$$

For any $\mathbf{B} \in \mathcal{V}^2$ we obtain

$$\begin{aligned} D\boldsymbol{\Sigma}(\mathbf{A})\mathbf{B} &= [D\boldsymbol{\Sigma}(\mathbf{A})]_{ijkl} B_{kl} \mathbf{e}_i \otimes \mathbf{e}_j \\ &= (\delta_{kl} A_{ij} B_{kl} + A_{mm} \delta_{ik} \delta_{jl} B_{kl}) \mathbf{e}_i \otimes \mathbf{e}_j \\ &= (A_{ij} B_{kk} + A_{mm} B_{ij}) \mathbf{e}_i \otimes \mathbf{e}_j \\ &= \text{tr}(\mathbf{B})\mathbf{A} + \text{tr}(\mathbf{A})\mathbf{B}. \end{aligned}$$

□

Bibliographic Notes

Much of the material on vector and tensor analysis in Cartesian coordinates can be found in standard texts on advanced multivariable calculus, for example Bourne and Kendall (1992) and Marsden and Tromba (1988). The book by Chadwick (1976) also contains a concise and readable treatment of this subject, oriented towards applications in continuum mechanics. For classic treatments in general curvilinear coordinates the reader is referred to Brand (1947), Sokolnikoff (1951) and Brillouin (1964). More modern treatments with applications can be found in Ogden (1984) and Simmonds (1994).

Proofs of the Divergence and Stokes' Theorems in general dimensions can be found in Munkres (1991). For an extensive analysis of the types of regions and surfaces for which these theorems are valid the reader is referred to Kellogg (1967).

Exercises

- 2.1 Consider the scalar field $\phi(\mathbf{x}) = (x_1)^2 x_3 + x_2 (x_3)^2$ and the vector field $\mathbf{v}(\mathbf{x}) = x_3 \mathbf{e}_1 + x_2 \sin(x_1) \mathbf{e}_3$. Find the components of $\nabla \phi(\mathbf{x})$ and $\nabla \mathbf{v}(\mathbf{x})$.

- 2.2 Consider the vector and second-order tensor fields

$$\mathbf{v}(\mathbf{x}) = x_1 \mathbf{e}_1 + x_2 x_1 \mathbf{e}_2 + x_3 x_1 \mathbf{e}_3,$$

$$\mathbf{S}(\mathbf{x}) = x_2 \mathbf{e}_1 \otimes \mathbf{e}_2 + x_1 x_3 \mathbf{e}_3 \otimes \mathbf{e}_3.$$

Find: (a) $\nabla \cdot \mathbf{v}$, (b) $\nabla \times \mathbf{v}$, (c) $\nabla \cdot \mathbf{S}$.

- 2.3 Use index notation to prove the following, where $\mathbf{A}$ is a constant second-order tensor:

(a) $\nabla \mathbf{x} = \mathbf{I}$,

(b) $\nabla \cdot \mathbf{x} = 3$,

(c) $\nabla(\mathbf{x} \cdot \mathbf{A} \mathbf{x}) = (\mathbf{A} + \mathbf{A}^T) \mathbf{x}$.

- 2.4 Let ϕ be a scalar field, $\mathbf{v}$ and $\mathbf{w}$ vector fields, and $\mathbf{S}$ a second-order tensor field. Use index notation to prove the following:

(a) $\nabla \cdot (\phi \mathbf{v}) = (\nabla \phi) \cdot \mathbf{v} + \phi(\nabla \cdot \mathbf{v})$,

(b) $\nabla(\mathbf{v} \cdot \mathbf{w}) = (\nabla \mathbf{v})^T \mathbf{w} + (\nabla \mathbf{w})^T \mathbf{v}$,

(c) $\nabla \cdot (\mathbf{v} \otimes \mathbf{w}) = (\nabla \mathbf{v}) \mathbf{w} + (\nabla \cdot \mathbf{w}) \mathbf{v}$.

- 2.5 Consider the scalar field $\phi(\mathbf{x}) = 1/|\mathbf{x}|$, $\mathbf{x} \neq \mathbf{0}$, and the vector field $\mathbf{v}(\mathbf{x}) = \phi(\mathbf{x}) \mathbf{n}$, where $\mathbf{n}$ is a constant vector. Show that:

(a) $\nabla \phi(\mathbf{x}) = -\mathbf{x}/|\mathbf{x}|^3$ for all $\mathbf{x} \neq \mathbf{0}$,

(b) $\Delta \phi(\mathbf{x}) = 0$ for all $\mathbf{x} \neq \mathbf{0}$,

(c) $\Delta \mathbf{v}(\mathbf{x}) = \mathbf{0}$ for all $\mathbf{x} \neq \mathbf{0}$.

Remark: Scalar and vector fields satisfying $\Delta \phi = 0$ and $\Delta \mathbf{v} = \mathbf{0}$ in a region B are said to be **harmonic** in B . In this sense the above fields are harmonic in any region which excludes the origin.

- 2.6 Consider a vector field $\mathbf{v} : \mathbf{E}^3 \rightarrow \mathcal{V}$. Show that:

(a) $\nabla \cdot (\nabla \mathbf{v}^T) = \nabla(\nabla \cdot \mathbf{v})$,

(b) $\nabla \times \mathbf{v} = \mathbf{0}$ if and only if $\nabla \mathbf{v} = \nabla \mathbf{v}^T$,

(c) if $\nabla \cdot \mathbf{v} = 0$ and $\nabla \times \mathbf{v} = \mathbf{0}$, then $\mathbf{v}$ is harmonic.

- 2.7 Let $\mathbf{v}$ and $\mathbf{w}$ be vector fields and ϕ a scalar field. Prove the following identities:

(a) $\nabla \cdot (\nabla \times \mathbf{v}) = 0$,

- (b) $\nabla \cdot (\mathbf{v} \times \mathbf{w}) = \mathbf{w} \cdot (\nabla \times \mathbf{v}) - \mathbf{v} \cdot (\nabla \times \mathbf{w})$,
 (c) $\nabla \times (\nabla \times \mathbf{v}) = \nabla(\nabla \cdot \mathbf{v}) - \Delta \mathbf{v}$,
 (d) $\nabla \times (\nabla \phi) = \mathbf{0}$.

- 2.8 Let B be a region in $\mathbf{E}^3$ with boundary ∂B , let $\mathbf{n}$ be the outward unit normal field on ∂B , and let $\mathbf{v}$ be a vector field in B . Use the divergence theorem for second-order tensors to show

$$\int_B \nabla \mathbf{v} \, dV = \int_{\partial B} \mathbf{v} \otimes \mathbf{n} \, dA.$$

Hint: Two second-order tensors $\mathbf{A}$ and $\mathbf{B}$ are equal if and only if $\mathbf{A}\mathbf{a} = \mathbf{B}\mathbf{a}$ for all vectors $\mathbf{a}$.

- 2.9 Let $\mathbf{S}$ be an arbitrary second-order tensor field and consider the vector field $\mathbf{x}$. Show that

$$\nabla \cdot (\mathbf{S}\mathbf{x}) = (\nabla \cdot \mathbf{S}^T) \cdot \mathbf{x} + \text{tr}(\mathbf{S}).$$

- 2.10 Let B be a region in $\mathbf{E}^3$ with boundary ∂B , and let $\mathbf{n}$ be the unit outward normal field on ∂B . Let $\mathbf{v}$ and $\mathbf{w}$ be vector fields and $\mathbf{S}$ a second-order tensor field in B . Use the divergence theorems to show:

- (a) $\int_{\partial B} (\mathbf{S}\mathbf{n}) \otimes \mathbf{v} \, dA = \int_B (\nabla \cdot \mathbf{S}) \otimes \mathbf{v} + \mathbf{S} \nabla \mathbf{v}^T \, dV$,
 (b) $\int_{\partial B} \mathbf{v} \cdot (\mathbf{S}\mathbf{n}) \, dA = \int_B (\nabla \cdot \mathbf{S}) \cdot \mathbf{v} + \mathbf{S} : \nabla \mathbf{v} \, dV$,
 (c) $\int_{\partial B} \mathbf{v}(\mathbf{w} \cdot \mathbf{n}) \, dA = \int_B (\nabla \cdot \mathbf{w})\mathbf{v} + (\nabla \mathbf{v})\mathbf{w} \, dV$.

- 2.11 Let ϕ , $\mathbf{v}$ and $\mathbf{S}$ be scalar, vector and second-order tensor fields, respectively. Prove the following identities:

- (a) $\nabla \cdot (\phi \mathbf{S}\mathbf{v}) = \phi(\nabla \cdot \mathbf{S}^T) \cdot \mathbf{v} + \nabla \phi \cdot (\mathbf{S}\mathbf{v}) + \phi \mathbf{S} : (\nabla \mathbf{v})^T$,
 (b) $\nabla \cdot (\mathbf{S}\mathbf{v}) = \mathbf{S}^T : \nabla \mathbf{v} + \mathbf{v} \cdot (\nabla \cdot \mathbf{S}^T)$.

- 2.12 Let $\mathbf{v}$ and $\mathbf{w}$ be time-dependent vector fields and consider the derivative operation defined by

$$\frac{D_{\mathbf{v}} \mathbf{w}}{Dt} = \frac{\partial \mathbf{w}}{\partial t} + (\nabla \cdot \mathbf{w})\mathbf{v} + \nabla \times (\mathbf{w} \times \mathbf{v}).$$

- (a) Show that

$$\frac{D_{\mathbf{v}} \mathbf{w}}{Dt} = \frac{\partial \mathbf{w}}{\partial t} + (\nabla \mathbf{w})\mathbf{v} - (\nabla \mathbf{v})\mathbf{w} + (\nabla \cdot \mathbf{v})\mathbf{w}.$$

(b) For any vector $\mathbf{b}$, show that

$$\mathbf{b} \cdot \frac{D\mathbf{w}}{Dt} = \mathbf{b} \cdot \left[\frac{\partial \mathbf{w}}{\partial t} + (\nabla \mathbf{w})\mathbf{v} \right] + [(\mathbf{b} \cdot \mathbf{w})\mathbf{I} - \mathbf{b} \otimes \mathbf{w}] : \nabla \mathbf{v}.$$

2.13 Let $\{\mathbf{e}_i\}$ and $\{\widehat{\mathbf{e}}_i(t)\}$ be two coordinate frames for $\mathbf{E}^3$, where the first is fixed (time-independent) and the second is moving (time-dependent). Moreover, let $\mathbf{Q}(t)$ be the change of basis tensor from $\{\mathbf{e}_i\}$ to $\{\widehat{\mathbf{e}}_i(t)\}$ so that (omitting arguments t for brevity)

$$\widehat{\mathbf{e}}_i = \mathbf{Q}\mathbf{e}_i.$$

(a) Show that the time derivative $d\widehat{\mathbf{e}}_i/dt$, as measured by an observer in the fixed frame, may be expressed as

$$\frac{d\widehat{\mathbf{e}}_i}{dt} = \boldsymbol{\Omega}\widehat{\mathbf{e}}_i = \boldsymbol{\omega} \times \widehat{\mathbf{e}}_i,$$

where $\boldsymbol{\Omega}(t)$ is the skew-symmetric tensor defined by

$$\frac{d\mathbf{Q}}{dt} = \boldsymbol{\Omega}\mathbf{Q} \quad \text{or equivalently} \quad \boldsymbol{\Omega} = \frac{d\mathbf{Q}}{dt}\mathbf{Q}^T,$$

and $\boldsymbol{\omega}(t)$ is the axial vector of $\boldsymbol{\Omega}(t)$.

(b) For any vector $\mathbf{v} = \widehat{v}_i\widehat{\mathbf{e}}_i = v_i\mathbf{e}_i$, show that

$$\frac{d\widehat{v}_i}{dt} = Q_{ji} \left[\frac{dv_j}{dt} - \Omega_{jk}v_k \right], \quad (2.3)$$

where Q_{ij} and Ω_{jk} are the components of $\mathbf{Q}$ and $\boldsymbol{\Omega}$ in the fixed frame.

Remark: An observer in the fixed frame $\{\mathbf{e}_i\}$ sees the time derivative of $\mathbf{v}$ as

$$\dot{\mathbf{v}} = \frac{dv_i}{dt}\mathbf{e}_i,$$

while an observer moving with the frame $\{\widehat{\mathbf{e}}_i(t)\}$ sees the time derivative of $\mathbf{v}$ as

$$\overset{\circ}{\mathbf{v}} = \frac{d\widehat{v}_j}{dt}\widehat{\mathbf{e}}_j.$$

The components of $\dot{\mathbf{v}}$ and $\overset{\circ}{\mathbf{v}}$ in the fixed frame $\{\mathbf{e}_i\}$ are

$$\begin{aligned} \dot{v}_i &= \dot{\mathbf{v}} \cdot \mathbf{e}_i = \frac{dv_i}{dt}, \\ \overset{\circ}{v}_i &= \overset{\circ}{\mathbf{v}} \cdot \mathbf{e}_i = \frac{d\widehat{v}_j}{dt}\widehat{\mathbf{e}}_j \cdot \mathbf{e}_i = Q_{ij}\frac{d\widehat{v}_j}{dt}. \end{aligned}$$

In view of (2.3) the vector $\overset{\circ}{\mathbf{v}}$ can be expressed in the fixed frame as

$$\overset{\circ}{\mathbf{v}} = \overset{\circ}{v}_i \mathbf{e}_i = \left[\frac{dv_i}{dt} - \Omega_{ik} v_k \right] \mathbf{e}_i.$$

The vector $\overset{\circ}{\mathbf{v}}$ is called the **co-rotational** or **Jaumann derivative** of $\mathbf{v}$ with respect to $\boldsymbol{\Omega}$. It represents the rate of change of $\mathbf{v}$ with respect to time as seen by an observer in a moving frame. In tensor notation we have

$$\overset{\circ}{\mathbf{v}} = \dot{\mathbf{v}} - \boldsymbol{\Omega} \mathbf{v}.$$

- 2.14 Let $\{\mathbf{e}_i\}$, $\{\hat{\mathbf{e}}_i(t)\}$, $\mathbf{Q}(t)$ and $\boldsymbol{\Omega}(t)$ be as in Exercise 13, and consider a second-order tensor $\mathbf{S} = S_{ij} \mathbf{e}_i \otimes \mathbf{e}_j = \hat{S}_{ij} \hat{\mathbf{e}}_i \otimes \hat{\mathbf{e}}_j$. Show that

$$\frac{d\hat{S}_{ij}}{dt} = Q_{ki} Q_{lj} \left[\frac{dS_{kl}}{dt} - \Omega_{km} S_{ml} - \Omega_{lm} S_{km} \right],$$

where Q_{ij} and Ω_{jk} are the components of $\mathbf{Q}$ and $\boldsymbol{\Omega}$ in the fixed frame $\{\mathbf{e}_i\}$.

Remark: The tensor $\overset{\circ}{\mathbf{S}}$ given by

$$\overset{\circ}{\mathbf{S}} = \left[\frac{dS_{kl}}{dt} - \Omega_{km} S_{ml} - \Omega_{lm} S_{km} \right] \mathbf{e}_k \otimes \mathbf{e}_l$$

is called the **co-rotational** or **Jaumann derivative** of $\mathbf{S}$ with respect to $\boldsymbol{\Omega}$. It represents the rate of change of $\mathbf{S}$ as seen by an observer in a moving frame. In tensor notation, noting that $\boldsymbol{\Omega}^T = -\boldsymbol{\Omega}$, we have

$$\overset{\circ}{\mathbf{S}} = \dot{\mathbf{S}} - \boldsymbol{\Omega} \mathbf{S} + \mathbf{S} \boldsymbol{\Omega}.$$

- 2.15 Consider the scalar-valued function $\psi(\mathbf{A}) = \text{tr}(\mathbf{A}) \mathbf{A} : \mathbf{B}$, where $\mathbf{B}$ is a constant second-order tensor. Show that

$$D\psi(\mathbf{A}) = (\mathbf{A} : \mathbf{B}) \mathbf{I} + \text{tr}(\mathbf{A}) \mathbf{B}.$$

- 2.16 Consider the tensor-valued function $\boldsymbol{\Sigma}(\mathbf{A}) = \mathbf{A}^2$. Show that

$$D\boldsymbol{\Sigma}(\mathbf{A})\mathbf{B} = \mathbf{B}\mathbf{A} + \mathbf{A}\mathbf{B}, \quad \forall \mathbf{B} \in \mathcal{V}^2.$$

- 2.17 Let $\mathbf{C} : \mathcal{V}^2 \rightarrow \mathcal{V}^2$ be a fourth-order tensor defined by $\mathbf{C}(\mathbf{A}) = \text{tr}(\mathbf{A})\mathbf{I} + \mathbf{A}$, and let $\psi : \mathcal{V}^2 \rightarrow \mathbb{R}$ be a function defined by $\psi(\mathbf{A}) = \frac{1}{2}\mathbf{A} : \mathbf{C}(\mathbf{A})$. Show that:

(a) $\mathbf{A} : \mathbf{C}(\mathbf{B}) = \mathbf{C}(\mathbf{A}) : \mathbf{B}, \quad \forall \mathbf{A}, \mathbf{B} \in \mathcal{V}^2,$

(b) $D\psi(\mathbf{A}) = \mathbf{C}(\mathbf{A}).$

Answers to Selected Exercises

2.1

$$[\nabla\phi] = \left\{ \begin{array}{c} \phi_{,1} \\ \phi_{,2} \\ \phi_{,3} \end{array} \right\} = \left\{ \begin{array}{c} 2x_1x_3 \\ (x_3)^2 \\ (x_1)^2 + 2x_2x_3 \end{array} \right\}.$$

$$[\nabla\mathbf{v}] = \begin{pmatrix} v_{1,1} & v_{1,2} & v_{1,3} \\ v_{2,1} & v_{2,2} & v_{2,3} \\ v_{3,1} & v_{3,2} & v_{3,3} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ x_2 \cos(x_1) & \sin(x_1) & 0 \end{pmatrix}.$$

2.3 (a) $\nabla \mathbf{x} = \frac{\partial x_i}{\partial x_j} \mathbf{e}_i \otimes \mathbf{e}_j = \delta_{ij} \mathbf{e}_i \otimes \mathbf{e}_j = \mathbf{I}.$

(b) $\nabla \cdot \mathbf{x} = \frac{\partial x_i}{\partial x_i} = \delta_{ii} = 3.$

(c)

$$\begin{aligned} \nabla(\mathbf{x} \cdot \mathbf{A}\mathbf{x}) &= \frac{\partial}{\partial x_i} (x_k A_{kl} x_l) \mathbf{e}_i \\ &= (\delta_{ki} A_{kl} x_l + x_k A_{kl} \delta_{li}) \mathbf{e}_i \\ &= (A_{il} x_l + A_{ki} x_k) \mathbf{e}_i \\ &= (\mathbf{A} + \mathbf{A}^T) \mathbf{x}. \end{aligned}$$

- 2.5 (a) Since $|\mathbf{x}| = (x_k x_k)^{1/2}$ and $\partial x_k / \partial x_i = \delta_{ki}$, we find by the chain rule that $\partial |\mathbf{x}| / \partial x_i = x_i / |\mathbf{x}|$. Thus

$$\nabla\phi = \frac{\partial}{\partial x_i} \left(\frac{1}{|\mathbf{x}|} \right) \mathbf{e}_i = -\frac{1}{|\mathbf{x}|^2} \frac{\partial |\mathbf{x}|}{\partial x_i} \mathbf{e}_i = -\frac{1}{|\mathbf{x}|^2} \frac{x_i}{|\mathbf{x}|} \mathbf{e}_i = -\frac{\mathbf{x}}{|\mathbf{x}|^3}.$$

- (b) Using the facts that $\Delta\phi = \nabla \cdot (\nabla\phi)$ and $[\nabla\phi]_i = -x_i/|\mathbf{x}|^3$,